Ankit

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Education

- **Indian Institute of Technology, Ropar** *2020 – 2024*
  Bachelor of Technology, Mathematics and Computing CGPA: 7.7/10

Experience

- **Futures First, Bangalore** *Aug 2024 – Present*
  Analyst
  - Traded **SOFR** and **Fed Funds (ZQ)** futures, focusing on multi-leg structures such as Calendar spreads, butterflies and double butterflies to manage risk and exposure across the interest rate curve.
  - Utilized and configured OMS execution modules **(RangeTrader, QuickASE, and ECI)** to apply rule-based execution logic for different strategies, ensuring deterministic behavior in real-time market conditions.
  - Analyzed macroeconomic indicators including **CPI, NFP, GDP, and unemployment data**, and executed event-driven decision logic based on expected interest-rate futures impacts under tight timing constraints.

- **Samsung, Bangalore** *Jun 2023 – Jul 2023*
  SDE Intern **Certificate**
  - Developed a semi-supervised **anomaly detection** system for large-scale log data using deep learning models, improving detection of system irregularities with minimal labeled input.
  - Designed and integrated **RESTful API** endpoints for session management (create, save, retrieve, list) into an internal tool, streamlining user interaction and enhancing backend functionality.

Projects

- **Macroeconomic Event Impact Analyzer and SOFR Intelligence Dashboard** *Oct 2025 – Present*
  Python, FastAPI, React, SQLite, FRED API, Google Gemini API **Github** **Live-Demo**
  - Built a full-stack SOFR rate dashboard using FastAPI, React, and SQLite that visualizes live forward curve movements, computes spreads and butterflies rate structures, and tracks basis point shifts around macro events (CPI, NFP, GDP) using real-time FRED API data, deployed live on Render + Vercel.
  - Integrated Google Gemini API to generate AI-powered summary for any macro release providing hawkish/dovish interpretation, curve movement analysis, and Fed rate path commentary – reducing manual post-event analysis time significantly.

- **Credit Risk Analysis** *Jan 2024 – May 2024*
  Python, Pandas, NumPy, Scikit-learn **Github**
  - Built a Python data pipeline to analyze company financial data and predict credit risk using **Logistic Regression, Random Forest, SVM, CNN and LSTM**; identified MScore as the most important feature and achieved 84% accuracy.
  - Implemented portfolio optimization using **Markowitz Mean Variance and Black-Litterman Allocation** models and developed a web interface to visualize risk, returns and optimized portfolio allocation.

- **Real-Time Face Recognition System** *Mar 2023 – Apr 2023*
  Python, HTML/CSS, OpenCV, CNN – Guided by Dr. GS Raju, IIT Ropar **Github**
  - Developed a web-based face recognition system for college attendance and gate entry/exit by integrating a live camera feed with a Python backend and **OpenCV**-based face detection pipeline.
  - Implemented real-time face preprocessing and **CNN**-based recognition (multi-face detection, resizing, normalization, grayscale optimization) to generate embeddings per frame and automate attendance logging.

Technical Skills

- **Programming Languages**: C/C++, Python, HTML/CSS, SQL
- **Libraries and Tools**: Git, PostgreSQL, NumPy, Pandas, Matplotlib, REST APIs, Scikit-learn, TT, ADL
- **Frameworks and Platforms**: FastAPI, React.js, OpenCV, Render, Vercel, SQLite

Key Courses

- **Computer Science**: Data Structures, Algorithms, Object-Oriented Programming, Databases, Operating Systems
- **Mathematics**: Probability and Statistics, Data Science, Mathematical Image Processing, Calculus and Linear Algebra

Achievements

- **Top 5% – Google CodeJam, Google KickStart and Facebook Hacker Cup**, *2022*
- **Competitive Programming**, Solved 600+ problems on LeetCode; rated on Codeforces & CodeChef *Profile*
- **1st Position, Maths Quiz – Cynosure**, National Symposium on Advances in Mathematics *Certificate*
- **Technical Workshops, Talks and Army Exhibition Head**, Managed a team of 30 and Organized Technical workshops on AI/ML & AR-VR, Lecture Talks , Army Exhibitions during our Annual Techno-Cultural Fest. *2023*